

# SANGJUN LEE

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## EDUCATION

**University of California, Berkeley** - *B.S. in Electrical Engineering and Computer Science*

**May 2026**

**Relevant Coursework:** Data Structures, Efficient Algorithms and Intractable Problems, Great Ideas of Computer Architecture, Designing Information Devices and Systems I, Discrete Mathematics and Probability Theory, Multivariable Calculus, Structure and Interpretation of Computer Programs, Foundations of Data Science

## SKILLS

**Programming Languages:** Python, Java, C, JavaScript, HTML/CSS, Swift, SQL

**Frameworks & Tools:** React, ReactNative, Node.js, Express.js, Flutter, FastAPI, SpringBoot, Firebase, Google Cloud Platform, Docker, PyTorch, scikit-learn, NumPy, OpenMPI, SIMD, RISC-V, Git, Linux

**Languages:** English, Spanish, Korean

## EXPERIENCE

**Korean Airforce Headquarters**

**Oct 2023 - May 2025** | *South Korea*

**Automated Runway Crack Detection System** | *Backend Developer*

**Nov 2024 - Current**

- Developed and optimized the backend infrastructure for an automated crack detection system for runways using Spring Boot, reducing runway inspection and labeling time from over 100 hours to 2-3 hours
- Designed and maintained an Oracle database to store over 50,000 runway slabs and their corresponding metadata
- Implemented robust authentication systems to manage user permissions for initiating crack detection models and viewing results
- Developed an AI server using FastAPI to integrate YOLO for crack detection, preprocess images, and manage seamless transfer of processed data between the AI server and image server

**Speech Diarization** | *AI and Backend Developer*

**Oct 2023 - Jul 2024**

- Trained and fine-tuned transcription and diarization models, cutting the time for manually transcribing meetings from weeks to minutes by automating the process with AI via a FastAPI server, serving over 100 daily users
- Optimized load balancing for high-performance tasks, improving the ability to process multiple transcription requests concurrently
- Developed in a high-security, closed network environment, implementing Docker for containerization and utilizing tmux along with other infrastructure tools to efficiently manage long-running processes and system resources
- Awarded the Prime Minister Award in Korea as part of a AI project that reduced the time soldiers spent on administrative tasks

**AlectoAI** | *Software Engineering Intern*

**May 2023 - Jul 2023**

- Developed robust and scalable APIs in Swift for a platform that detects image misuse through facial recognition AI
- Created new functionalities such as commenting and liking on posts by connecting backend logic to existing frontend elements
- Managed the Firebase and constructed the schema to ensure efficient data storage and retrieval while maintaining performance

**KidNCrew** | *React Native Developer*

**Jan 2023 - June 2023** | *Berkeley, CA*

- Created intuitive user interfaces for a sports social media app using React Native, enhancing the overall user experience
- Integrated frontend features with the Supabase backend and used sockets to enable real-time chat and comment functionality

## PROJECTS

**Rook** — *Full Stack Developer*

**Oct 2024 - Current**

- Developed a full-stack web application using React and Express that recommends similar books based on a user's reading list
- Applied prompt engineering with OpenAI to generate book recommendations and used the Google Books API for book metadata
- Built a user authentication system with login/logout functionality and authorization with JWT to ensure protected user connection

**Gitlet**

**May 2022**

- Developed a version-control system that mimics Git's core functionalities, including initialization, staging, committing, and more
- Leveraged serialization and hashing techniques to efficiently save files and commit trees, ensuring persistence across versions
- Implemented advanced tree traversal algorithms to optimize performance while navigating through commit histories and branches

**RentMe** — *Full Stack Developer*

**Aug 2021 - Dec 2021**

- Collaborated with a team and built a full-stack mobile application in Swift enabling college students to rent out sports equipment
- Designed and implemented a responsive user interface using Auto Layout and UIKit frameworks for essential functionalities

## ACTIVITIES

**Teacher Assistant for Great Ideas in Computer Architecture (Machine Structures)**

**June 2023 - Aug 2023**

- Created weekly discussion worksheets for over 200 students and led lessons, covering key concepts such as C, RISC-V, hardware, and optimization (SIMD, OpenMP, OpenMP)
- Held office hours to assist students with debugging and assignments and graded exams to assess understanding of course material